

Business Context & Problem Definition

Business Context

Medical imaging is a foundational capability in modern healthcare, particularly in cardiology, oncology, neurology, and emergency medicine. Hospitals and diagnostic centers rely on imaging modalities such as CT, MRI, and angiography to support diagnosis, treatment planning, and surgical preparation.

In large healthcare ecosystems, imaging data is typically managed through Picture Archiving and Communication Systems (PACS) and related hospital information systems. These environments are characterized by:

- High throughput of imaging studies (continuous inflow of DICOM series)
- Distributed infrastructure across departments, hospitals, or regional networks
- Strict regulatory constraints (GDPR in EU, medical data governance rules)
- Strong dependency on specialist interpretation (radiologists, cardiologists)

Despite digitization, the clinical workflow remains largely human-centric and sequential, where images are acquired, stored, retrieved, and manually interpreted by specialists under time pressure.

At the same time, AI technologies (especially deep learning models for segmentation, classification, and reconstruction) have reached a level of maturity where they can significantly augment diagnostic workflows. However, their adoption in production environments is still limited due to lack of orchestration, integration complexity, and governance concerns.

Current AS-IS Process

A simplified representation of the current workflow:

1. Image Acquisition
 - CT/MRI scanners generate DICOM series
 - Data is sent to PACS or local hospital storage
2. Storage & Retrieval
 - Images stored in siloed systems
 - Retrieval depends on hospital workflows and user permissions
3. Manual Analysis
 - Radiologist loads DICOM series
 - Performs slice-by-slice inspection
 - Mentally reconstructs 3D anatomy
4. Report Generation
 - Findings are manually documented
 - Reports are stored in HIS/RIS systems
5. Optional AI Tools (fragmented)
 - Standalone tools may provide segmentation or detection
 - Outputs are not consistently integrated into workflow

Key Pain Points

1. Cognitive Load & 3D Reconstruction Complexity

- Radiologists must mentally reconstruct 3D structures from 2D slices
- High complexity in cardiac and vascular imaging
- Increased risk of inter-observer variability

2. Time Constraints & Operational Bottlenecks

- High volume of studies per day per radiologist
- Delays in urgent cases (e.g., stroke, cardiac events)
- Limited scalability of expert workforce

3. Fragmented AI Adoption

- AI tools are typically single-purpose (e.g., segmentation only)
- Lack of orchestration across multiple models
- No unified pipeline from ingestion → inference → reporting

4. Data Silos & Interoperability Issues

- PACS systems are vendor-specific and not easily interoperable
- Difficulty in aggregating data across institutions
- Limited cross-patient analytics capability

5. Regulatory & Privacy Constraints

- Strict GDPR compliance requirements for patient data
- Restrictions on external cloud processing in many institutions
- Need for auditability and traceability of AI decisions

6. Lack of Clinical Trust in AI Outputs

- AI outputs often perceived as “black box”
- No structured validation layer integrated into workflow
- Limited explainability and confidence scoring in production use

Core Problem Definition

The fundamental challenge is not merely the automation of image analysis, but the absence of a **cohesive orchestration layer that can integrate AI capabilities into a clinically safe, interpretable, and workflow-aligned system.**

More specifically:

How can we design a system capable of orchestrating multiple specialized AI models across distributed medical imaging environments, while preserving clinical reliability, regulatory compliance, and integration with existing hospital workflows?

This problem spans multiple dimensions:

- **Technical:** integration of heterogeneous AI models and data sources
- **Architectural:** need for distributed, hybrid (edge + cloud) processing
- **Operational:** alignment with real clinical workflows and time constraints
- **Regulatory:** strict compliance with healthcare data governance
- **Human factors:** ensuring trust, interpretability, and usability for clinicians

Strategic Gap

The key gap between current systems and the desired future state is:

- From isolated AI tools → to coordinated AI agent ecosystem
- From manual interpretation of 2D slices → to assisted 3D reasoning systems
- From static reporting pipelines → to dynamic, interactive decision support workflows